

### **Task 3: The ten components of the protocol**

#### **1. Title**

Sociotechnical strategy based on predictive analytics and regulatory governance for the prevention of tax fraud within SUNAT's Electronic Issuance System.

#### **2. Abstract**

The study develops a framework to prevent tax fraud involving fictitious transactions within SUNAT's Electronic Issuance System (SEE). This framework utilizes the technical core of a Geometric Deep Learning architecture based on Graph Neural Networks (GNN) to model XML transactions in real time, thereby detecting collusion networks and shell companies.

#### **3. Introduction and problem statement**

Since 2018, the use of electronic payment vouchers (SEE) has become widespread in Peru, and with this shift, tax fraud has grown increasingly sophisticated. Tax evaders employ automated methods to issue fake invoices through interconnected networks of shell companies, simulating transactions to claim improper tax credits. In this context, current audits are reactive and rely on linear rules that fail to detect complex, dynamic patterns. The use of AI has been characterized as involving "black-box" schemes that compromise tax confidentiality and create legal liabilities due to a lack of algorithmic transparency. Given this situation, how can a sociotechnical strategy be structured—integrating advanced predictive analytics and regulatory governance—to optimize the early detection of tax fraud within SUNAT's SEE system?

#### **4. Literature review**

Geometric Deep Learning (GNNs) in RegTech: transactions of tax interest are mathematically defined as a directed multi-relational graph  $G = (V, E)$ , where vertices ( $V$ ) represent taxpayers and edges ( $E$ ) represent electronic invoices. Algorithms such as Graph Attention Networks (GAT) and GraphSAGE enable the calculation of structural embeddings that identify cyclical structures and tax credit laundering schemes—patterns impossible to detect using traditional tabular machine learning.

Explainable AI (XAI) in Public Administration: the use of "black-box" models in the government sector compromises the principle of transparency. Local attribution methods like SHAP (SHapley Additive exPlanations) allow deep network predictions to be broken down into human-understandable variables, thereby lending legal validity to administrative actions.

Sociotechnical systems and tax compliance: in line with OECD and CIAT standards, compliance risk management should be conceived as a sociotechnical system in which technology (algorithms) and the social system (auditors, laws, taxpayers) co-evolve; while pure automation generates friction, a hybrid design (human-in-the-loop) maximizes regulatory effectiveness.

#### **5. Hypotheses and objectives**

Objective: To design a sociotechnical framework based on Geometric Deep Learning and RegTech governance for the proactive prevention of tax fraud within SUNAT's SEE (Electronic Billing System).

Hypothesis: A sociotechnical framework integrating explainable GNNs and proactive regulatory governance reduces the tax fraud rate in the SEE and minimizes legal contingencies, outperforming the efficiency of rule-based or black-box AI systems.

## **6. Methodology**

The methodological design is applied and experimental, employing a sequential mixed-methods approach. This involves data engineering and predictive modeling (QUAN); attribution calculation processes—such as KernelSHAP applied to graph embeddings—to extract the feature vectors that triggered the fraud alert; and regulatory mapping. Finally, the framework's regulatory and operational viability is validated through consultation with expert auditors, data engineers, and tax lawyers.

## **7. Ethical considerations**

The study requires respect for taxpayer confidentiality, the protection of algorithmic equity across economic sectors, and a guarantee of traceability regarding the variables that triggered each alert, enabling external audits of the algorithm.

## **8. Expected results**

A strengthened GNN-Tax architecture design, including source code and weights for a graph neural network optimized to detect financial collusion within tax-related Big Data. Additionally, the development of a legal explainability pipeline using software that transforms abstract mathematical metrics into readable technical reports to provide legal substantiation for tax adjustments. Finally, the design of a validated sociotechnical governance framework to enable the implementation of Deep Tech models within the tax administrations of emerging economies.

## **9. Schedule and budget**

## **10. Bibliography**

- Inter-American Center of Tax Administrations - CIAT (2020). *Artificial intelligence applied to tax auditing*.
- National Superintendence of Customs and Tax Administration - SUNAT (2024). *Digital Government Plan 2025–2027*. Lima.
- National Superintendence of Customs and Tax Administration - SUNAT (2023). *Compliance risk management in tax administration and estimation of the tax gap*. Revista de Tributación y Aduanas, 5(2), 45-62.
- National Superintendence of Customs and Tax Administration – SUNAT (2011). *Consolidated Text of the Tax Code – Supreme Decree No. 133-2013-EF and amendments*. SUNAT Legislation.